Federated Deep Learning for Alzheimer's Disease Detection: Strong Internal Performance Does Not Survive a Change of Imaging Source

Abstract

Deep learning can stage Alzheimer's disease from magnetic resonance imaging, but performance on held-out images from the training source may not generalize across imaging collections. We developed a dual-channel classifier combining ConvNeXt-Small and Swin-V2-S representations with Convolutional Block Attention Module fusion. The executed objective used soft-target cross-entropy and focal loss; a configured supervised contrastive term was numerically inactive. A 6,400-image four-class training collection lacked subject identifiers and was split at image level. We compared centralized training with federated averaging, federated proximal optimization and a federated batch-normalization-style strategy using two simulated same-source clients. Evaluation comprised 951 internal images and 86,437 OASIS-1-derived external images. Internally, the batch-normalization-style model achieved 92.74% accuracy, 95.13% balanced accuracy, a macro-F_1 score of 0.945 and a macro area under the receiver operating characteristic curve of 0.991. Externally, accuracy fell to 12.06–22.11%, macro-F_1 to 0.097–0.173 and macro area under the curve to 0.634–0.692. Raw source statistics differed substantially, with Kolmogorov–Smirnov statistics of 0.967 and 0.997 and $\mathrm{MMD}^2 = 0.534$. Temperature scaling increased validation expected calibration error from 0.348 to 0.390, and the batch-normalization-style model produced 9,711 high-confidence external errors compared with 10 internally. Strong same-source performance and simulated federated training therefore did not establish cross-source robustness. Without subject identifiers, internal performance must be interpreted as an image-level upper-bound estimate rather than evidence of subject-independent generalization.

Keywords: Alzheimer's disease, Magnetic resonance imaging, Federated deep learning, Distribution shift, Model calibration, Implementation audit

1 Introduction

Alzheimer's disease (AD) is the leading cause of dementia and accounts for approximately 60% to 80% of cases worldwide [1–5]. Its pathological processes may begin many years before clinical symptoms become apparent, with amyloid-beta accumulation, tau pathology and progressive tissue loss affecting memory-related structures

such as the hippocampus and entorhinal cortex [2, 3, 6–8]. Early identification is therefore important, particularly during mild cognitive impairment, when clinical and functional decline may still be limited. Cognitive instruments such as the Mini-Mental State Examination (MMSE) and Clinical Dementia Rating (CDR) remain central to assessment, but structural magnetic resonance imaging (MRI) provides an important non-invasive view of neuroanatomical change [9, 10].

Deep learning has expanded the capacity of MRI-based systems to learn subtle image representations without relying entirely on handcrafted features. Convolutional neural networks (CNNs) capture local texture and structural patterns, whereas vision transformers model relationships across more distant image regions. Hybrid CNN–transformer models have consequently reported strong results on public Alzheimer imaging benchmarks [2, 11–13]. Their clinical translation nevertheless remains constrained by privacy, interpretability and generalization. Federated learning addresses the first concern by allowing institutions to train a shared model without exchanging the underlying scans [7, 14, 15], while attribution methods seek to clarify which image regions influence individual predictions [16–18].

Generalization across imaging sources is a particularly important and sometimes underexamined limitation. Many published results are obtained from a held-out partition of the same collection used for training. Such evaluation may preserve source-specific acquisition, preprocessing and sampling characteristics and may also be vulnerable to image-level leakage when subject identifiers are unavailable. Reviews have reported a marked difference between internal and external performance and have cautioned that near-perfect results may reflect single-dataset evaluation rather than robust clinical discrimination [19, 20]. This concern is especially relevant to federated learning, whose intended use involves institutions that may differ in scanner characteristics, image processing and patient composition.

To examine this problem, we developed a dual-channel framework that combines a ConvNeXt-Small convolutional branch with a Swin-V2-S transformer branch. Their pooled representations are concatenated and refined using a Convolutional Block Attention Module (CBAM), followed by a projection and classification head. Training used soft-target cross-entropy and focal loss, with hyperparameters selected through Optuna; a configured supervised contrastive component was rendered inactive by the executed numerical guard. We evaluated centralized training alongside federated averaging (FedAvg), federated proximal optimization (FedProx) and a federated batch-normalization-style (FedBN-style) rule. The federated experiments used two simulated clients drawn from one 6,400-image source; they should therefore be interpreted as controlled aggregation experiments rather than a genuine multi-institutional federation. The resulting checkpoints were tested on a 951-image internal partition and on 86,437 images derived from the Open Access Series of Imaging Studies (OASIS-1).

The study was guided by four research questions:

RQ1 How do centralized, FedAvg, FedProx and FedBN-style training compare on the same-source internal test partition?

RQ2 Does the relative performance and ranking of these models persist when evaluation moves to a separately sourced imaging collection?

RQ3 How strongly do the two image sources differ in measurable intensity statistics, and how does cross-source degradation compare with robustness to synthetic corruption?

RQ4 What do calibration, high-confidence errors, statistical comparisons, qualitative attribution and computational cost reveal about the reliability and practical limitations of the framework?

Internal evaluation favoured the FedBN-style checkpoint, which achieved 95.13% balanced accuracy and a macro area under the receiver operating characteristic curve (macro-AUC) of 0.991. External evaluation led to a substantially different conclusion: accuracy fell to 12.06–22.11%, macro-AUC fell to 0.634–0.692, and the internal macro-AUC ranking did not match the external ranking. These results shifted the emphasis of the study from architectural performance alone to the reliability of the evaluation protocol.

The principal contributions of this work are as follows:

- We implement and evaluate a dual-channel ConvNeXt–Swin framework under centralized training and three two-client federated strategies using one consistent four-class evaluation pipeline, with additional single-backbone baselines and component ablations.
- We quantify the internal-to-external performance loss and report raw distributional differences between the training and OASIS-1-derived collections, while avoiding a causal interpretation that the available acquisition metadata cannot support.
- We examine reliability beyond discrimination through calibration, confidence-based error analysis, synthetic corruption, paired statistical testing, implementation-diagnostic attribution maps and a computational cost assessment.
- We distinguish completed empirical evidence from proxy, simulated, unsuccessful or unavailable analyses and explicitly acknowledge that the training collection lacks subject identifiers. The internal results must therefore be treated as image-level upper-bound estimates rather than evidence of subject-independent performance.

2 Related Work

Research on automated Alzheimer's disease assessment has progressed from conventional convolutional networks to multimodal, longitudinal and hybrid attention-based models. Explainability and privacy-preserving learning have also received increasing attention. Although many studies report high predictive performance, their results are difficult to compare directly because the imaging modality, classification task, dataset construction and validation protocol vary substantially. We reviewed forty studies across six dimensions relevant to the present work: hybrid modelling, hyperparameter optimization, external evaluation, federated learning, explainability and calibration. Tables 1 and 2 provide a structured comparison, and the following discussion identifies the methodological patterns and unresolved issues that informed our study.

2.1 From CNNs to Hybrid Attention Architectures

Earlier computer-aided systems relied largely on handcrafted measurements from selected anatomical regions, including hippocampal volume. Deep neural networks reduced this dependence by learning hierarchical representations directly from imaging data [1, 2]. Reviews covering ADNI, OASIS, AIBL and Kaggle-derived collections report accuracies of approximately 90–99%, although these figures combine different modalities, class definitions and validation procedures [1, 21, 22]. For example, Helaly et al. [2] evaluated VGG19 and a 3D CNN and reported accuracies of 97% and 95.17%, respectively. Ahila et al. [23] achieved 96.8% using FDG-PET, while Fathi et al. [3] reported 98.57% for binary classification and 93.88% for four-class staging using a six-model CNN ensemble. Further studies reported 98.67% with fine-tuned InceptionV3 [24], 99.68% with a regularized CNN on OASIS [25], and 99.4% with a lightweight network containing 1.2 million parameters [26]. These findings demonstrate the representational capacity of CNNs; however, differences in reported accuracy cannot be attributed to architecture alone because the underlying datasets, prediction tasks and evaluation designs are not equivalent.

More recent studies combine convolutional feature extraction with attention mechanisms or vision transformers. Kumar et al. [11] integrated a CNN with a Convolutional Block Attention Module and metaheuristic optimization, reporting 98.28% accuracy. A CNN–Swin-T architecture accompanied by explainability methods achieved 99.92% [12], while a ResNeXt101–Swin Transformer model achieved 98.83% [13]. Other configurations include a 2D ResNet-18 combined with a 3D CNN encoder [27] and EfficientNet combined with squeeze attention [28]. By contrast, Alsubaie et al. [29] evaluated a three-dimensional framework integrating multiscale feature fusion and a vision transformer on 4,706 ADNI scans under a subject-level split, obtaining 92.78% accuracy. Although this value is lower than several image-level results, it was obtained under a stricter independence criterion. Taken together, these studies support the use of complementary convolutional and transformer representations, while also demonstrating that performance must be interpreted in relation to dataset provenance and splitting strategy.

2.2 Explainability

Thirteen of the forty studies include an attribution or feature-importance component. Grad-CAM and Grad-CAM++ are used most frequently in image-based models [11–13, 28]; SHAP [30, 31] and LIME [15] appear less often. Zahedipour et al. [16] combined a genetic algorithm with Deep SHAP to reduce 96 anatomically defined regions to 11 influential regions. In a broader evaluation, Bloch et al. [18] compared SHAP, LIME, Grad-CAM, Grad-CAM++ and permutation importance across ADNI, AIBL and OASIS, using voxel-based morphometry as an anatomical reference. Models with similar predictive performance nevertheless emphasized different brain regions. In a related brain-imaging task, Yoon and Lin [17] also showed that LIME explanations could be influenced by irrelevant image edges and background. These results justify the use of multiple explanation methods, but they also caution against treating visually plausible heatmaps as evidence of anatomical validity. Quantitative faithfulness assessment,

anatomical registration and expert evaluation remain uncommon, and recent reviews continue to identify explainability as an unresolved challenge [7, 20, 32].

2.3 Federated Learning

Privacy-preserving learning is less extensively studied than model architecture. Toumaj et al. [7] found that 88.9% of the studies included in their review used no privacy mechanism, while other reviews present federated learning mainly as a direction for future research [32, 33]. Only two primary studies in Tables 1 and 2 implemented federated training. Abubakr et al. [14] combined EfficientNetB0, CBAM and FedAvg for binary classification, reporting 94.7% federated accuracy and 95.0% accuracy on an external image collection. Yao [15] distributed 6,400 Kaggle images among eight encrypted clients and reported 98.3% accuracy. These studies establish the feasibility of distributed optimization for Alzheimer's disease classification. However, their clients were created by partitioning public datasets rather than by using independently acquired institutional cohorts. They therefore did not directly test whether aggregation could accommodate scanner- or site-related heterogeneity, and neither study compared FedAvg with FedProx and a batch-normalization-oriented aggregation strategy.

2.4 Evaluation Practice and Generalizability

The central methodological question is not simply whether high accuracy can be achieved, but what the evaluation protocol permits that accuracy to establish. Reviews have warned that near-perfect results may be influenced by image-level splitting, augmentation before data partitioning or evaluation on a single dataset [20, 21, 34]. Several primary studies illustrate these concerns. Arafa et al. [6] reported 99.98–99.99% accuracy on an augmented 6,400-image Kaggle collection. Shamrat et al. [24] expanded their dataset approximately tenfold through augmentation without reporting a subject-level control. Akindele et al. [27] reported 99.99% accuracy and an area under the curve of 1.0 after constructing 3D inputs from augmented 2D images. Ullah et al. [26] explicitly used an image-level split because subject identifiers were unavailable. These results characterize performance on the reported partitions, but they do not establish independence at the patient level.

Evaluation on a second collection generally produces less optimistic results. Sharma et al. [8] reported a decline from 90.4% to 71.1% across two Kaggle datasets, while Zahedipour et al. [16] reported 96% on the training data and 85% on a 140-scan test collection. Similarly, Nibila and Sivagami [19] contrasted internal accuracies of 98–99% with external accuracies of 64–71%. These findings show that describing an experiment as "external validation" is not sufficient: the relationship between the development and evaluation sources, including their provenance and acquisition characteristics, must also be reported.

Probability calibration has received still less attention. Of the forty studies summarized in Tables 1 and 2, only Bloch et al. [18] report a calibration analysis. Most studies focus on accuracy, sensitivity, specificity, F_1 score and area under the curve, with some reporting near-perfect discrimination [11, 13]. For example, Balaji et al. [35] reported 98.5% accuracy together with an area under the curve of 0.80–0.83. These measures

describe different aspects of predictive performance, but none establishes whether the reported confidence values are reliable. The limited attention given to calibration is therefore important when models are proposed for clinical decision support.

2.5 Positioning of the Present Study

Four related gaps emerge from the literature. First, hybrid CNN–transformer systems are commonly evaluated on same-source partitions, and subject-level independence is not always documented. Second, evaluation on a distinctly sourced imaging collection remains uncommon and is rarely accompanied by a quantitative description of the difference between sources. Third, federated studies generally simulate clients by partitioning a single dataset and therefore do not reproduce the acquisition heterogeneity expected across institutions. Fourth, probability calibration and quantitative validation of model explanations are seldom reported.

The present study responds to these gaps while remaining explicit about the limitations of the available evidence. We compare centralized training with FedAvg, FedProx and a FedBN-style rule using a hybrid ConvNeXt–Swin model. Each strategy is evaluated on a same-source internal partition and a separately sourced OASIS-1-derived collection. We also measure raw differences between the image sources and examine validation-set calibration, confidence-stratified errors, synthetic corruption, attribution implementation diagnostics and computational cost. Because the training collection lacks subject identifiers and both simulated clients originate from that collection, we do not claim patient-level independence or genuine multi-institutional federation. The attribution outputs do not support an explainability claim because several routines depart from their canonical algorithms and anatomical registration, expert ratings and quantitative faithfulness tests were not completed. This deliberately bounded design allows the contrast between strong internal performance and poor cross-source performance to be reported without overstating the clinical, federated or interpretability evidence.

Table 1 Comparison of related studies 1–20. Hyb: hybrid architecture; HPO: hyperparameter optimization; Ext: external evaluation; FL: federated learning; XAI: explainable artificial intelligence; Cal: calibration. Y and N indicate whether a component was reported; – denotes not applicable. Reported acc. is the highest accuracy stated in each source and should not be compared directly across different tasks, datasets or validation protocols.

Study	Dataset/modality	Reported acc.	Hyb	HPO	Ext	FL	XAI	Cal
Gao & Lima (2022)	ADNI, OASIS, AIBL; review	98.8	–	–	–	N	N	N
Helaly (2022)	ADNI, 48k aug.	97.0	N	N	N	N	N	N
Ahila et al. (2022)	ADNI, 855 PET	96.8	N	N	N	N	N	N
Fathi (2024)	ADNI + local cohort	98.6	N	N	Y	N	N	N
Rao & Aparna (2023)	ADNI, OASIS, IBSR; review	99.0	–	–	–	N	N	N
Arafa (2024)	Kaggle AD MRI	99.99	N	N	N	N	N	N
Shamrat (2023)	ADNI, 60k aug.	98.7	N	N	N	N	N	N
Mmadumbu (2025)	NACC + ADNI	99.8	Y	N	N	N	N	N
Toumaj (2025)	Multiple; review	–	–	–	–	N	–	N
Mohsen (2025)	Multiple; review	99.9	–	–	–	N	–	N
Alsubaie (2024)	ADNI and others; review	99.6	–	–	–	N	–	N
Balaji (2023)	Kaggle MRI, Munich PET	98.5	Y	Y	N	N	N	N
Arya (2023)	ADNI, OASIS, Kaggle; review	99.4	–	–	–	N	–	N
Shaikh (2025)	ADNI, OASIS, NACC; review	98.2	–	–	–	N	–	N
Sharma (2022)	Two Kaggle MRI sets	90.4	N	N	Y	N	N	N
Zahedipour (2026)	ADNI, 140 scans	85.0	N	N	N	N	Y	N
Akindele (2026)	Kaggle + MIRIAD	99.99	Y	N	Y	N	N	N
Alsubaie (2026)	ADNI, 4,706 scans	92.8	Y	N	N	N	Y	N
Kumar (2026)	ADNI, 12.8k slices	98.3	Y	Y	N	N	Y	N
Hasan (2026)	606 microscopy	61.0	Y	N	N	N	N	N

Table 2 Comparison of related studies 21–40 and the present work. Column abbreviations and symbols follow Table 1.

Study	Dataset/modality	Reported acc.	Hyb	HPO	Ext	FL	XAI	Cal
Nibila & Sivagami (2026)	ADNI, OASIS, NACC; review	99.0	–	–	–	N	–	N
Mohsen (2026)	Kaggle MRI, 6,336	99.9	Y	N	N	N	Y	N
Resen (2026)	ADNI, OASIS, AIBL; review	98.0	–	–	–	N	–	N
Ullah (2026)	Kaggle MRI, 6,400	99.4	N	Y	N	N	N	N
Hariharan & Jothi (2026)	GSE63060/61, ADNI	91.0	N	N	N	N	Y	N
Mohsen (2026)	Kaggle MRI, 6,400	98.8	Y	Y	N	N	Y	N
Abubakr (2026)	Kaggle + ext. MRI	95.7	Y	N	Y	Y	Y	N
Yoon & Lin (2025)	Kaggle tumour MRI	99.0	N	N	N	N	Y	N
Song (2024)	ADNI, 268 fMRI	96.4	N	N	N	N	Y	N
Khanapur (2026)	Kaggle MRI, 6,400	91.7	N	N	N	N	Y	N
Yao (2026)	Kaggle MRI, 6,400	98.3	N	N	N	Y	Y	N
El-Geneedy et al. (2023)	OASIS MRI	99.7	N	N	N	N	N	N
Warren & Moustafa (2023)	ADNI, fMRI; review	97.0	–	–	–	N	–	N
Vrahatis (2023)	Non-invasive; review	–	–	–	–	N	–	N
Al Olaimat (2023)	ADNI + NACC	–	Y	N	Y	N	N	N
Asaduzzaman (2025)	Kaggle MRI, 6,400	97.3	N	N	N	N	N	N
Bloch (2024)	ADNI, AIBL, OASIS	–	N	N	Y	N	Y	Y
Abdelwahab (2023)	GSE5281/63060/61	97.1	N	N	N	N	N	N
Chua (2025)	OCT cohorts	–	Y	N	Y	N	N	N
Kina (2025)	Merged Kaggle MRI	–	Y	N	N	N	Y	N
This work	**Kaggle 6,400, OASIS 86k**	**92.7**	**Y**	**Y**	**Y**	**Y**	**Y**	**Y**

3 Methods

3.1 Overview

Figure 1 summarizes the complete three-part study workflow, from data preparation and model development through simulated federated training and internal–external evaluation. Two principles guided the evaluation. First, the same routine was applied to every checkpoint on both test sets, and the resulting predictions were used consistently for the metric, confusion-matrix, receiver-operating-characteristic and error analyses. Second, calibration was assessed separately using the centralized validation predictions. Because the framework is intended for use across institutions, the external collection provides the principal evidence of generalizability, while the internal partition serves as a same-source reference.

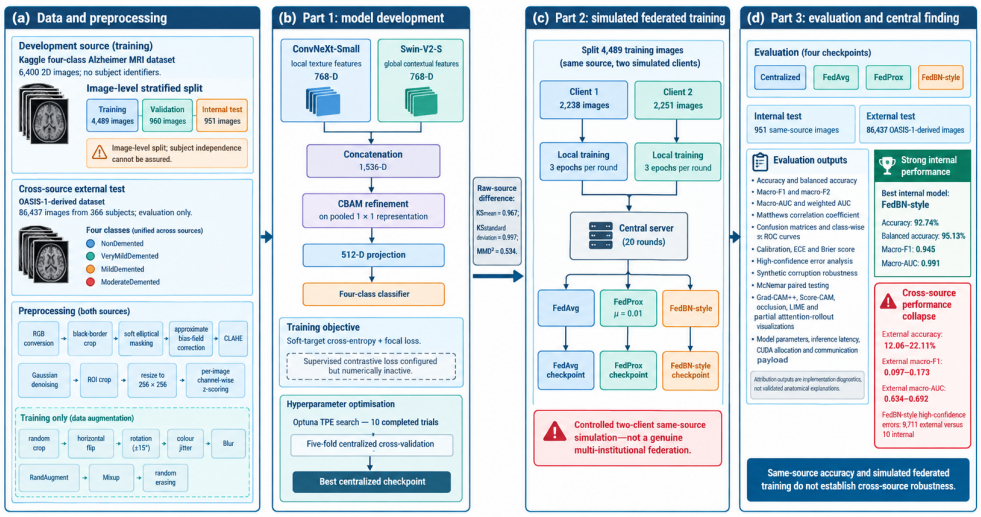


Fig. 1 Complete study workflow. (a) Development and cross-source data, splitting and preprocessing. (b) Dual-channel model development and centralized cross-validation. (c) Two-client same-source simulation using FedAvg, FedProx and a FedBN-style rule. (d) Internal and external evaluation. Strong internal performance did not persist on the OASIS-1-derived collection.

3.2 Cohorts and Evaluation Protocol

Two public collections were used (Table 3). The training cohort is the original, non-augmented four-class Alzheimer MRI collection distributed on Kaggle [41]: 6,400 axial slices comprising 3,200 NonDemented (50.0%), 2,240 VeryMildDemented (35.0%), 896 MildDemented (14.0%) and 64 ModerateDemented (1.0%), a majority-to-minority ratio of 50:1. A stratified 70/15/15 split with a fixed seed gave 4,489 training, 960 validation and 951 internal test images, the test partition containing 480, 332, 130

and 9 images in order of increasing severity. All training, validation, hyperparameter search and federated client data come from this cohort.

The external collection is a Kaggle derivative of OASIS-1 [42–44]: 86,437 slices from 366 subjects, comprising 67,222 NonDemented (77.8%), 13,725 VeryMildDemented (15.9%), 5,002 MildDemented (5.8%) and 488 ModerateDemented (0.6%), a ratio of 138:1. It contributed no training or validation data and was scored once per checkpoint. We call it a cross-dataset external test set rather than an independent patient cohort, because subject-level overlap with the training cohort has not been established: the Kaggle collection ships no provenance metadata and several public four-class releases are themselves OASIS-derived.

Two differences between the collections are particularly important when interpreting the external results. The external collection is approximately ninety times larger than the internal test partition, and its class distribution differs substantially, with the proportion of NonDemented images increasing from 50.0% to 77.8%. Consequently, reliance on the training-set class distribution may reduce external performance independently of any difference in image acquisition.

3.2.1 Splitting and the limits of the leakage control

OASIS filenames encode subject and session, and the pattern `OAS<n>_<n>_MR<n>` recovers 366 subjects. The Kaggle cohort ships no subject identifiers, so each file received a unique surrogate identifier, which makes grouping by identifier equivalent to an image-level split.

This limitation constrains the interpretation of the internal results. Because the Kaggle collection contains 2D images without subject identifiers, images from the same subject, if present, could not be assigned to a single partition. Subject-level or near-duplicate leakage therefore cannot be excluded or quantified. The internal results in Section 4.2 are accordingly treated as upper-bound estimates. OASIS was the only collection for which subject identifiers were available, and it contributed no training data.

Table 3 Image collections and their role in the evaluation protocol

Collection	Role	N images	Note
Kaggle four-class AD MRI	Training	4,489	Source of all federated clients
Kaggle four-class AD MRI	Validation	960	Early stopping and calibration
Kaggle four-class AD MRI	Internal test	951	Same source as training; no subject IDs
OASIS-1-derived	External test	86,437	366 subjects; overlap not established

Training cohort totals 6,400 images (3,200 / 2,240 / 896 / 64). Internal test class counts: 480 / 332 / 130 / 9. External: 67,222 / 13,725 / 5,002 / 488, in order of increasing severity.

3.3 Preprocessing and Augmentation

Both collections pass through the same deterministic preprocessing code; this standardizes the operations but does not guarantee that their resulting distributions match. Images are converted to three-channel RGB; black scanner borders are removed by thresholding at an intensity sum of 10 and cropping to the foreground bounding box with a five-pixel margin. A soft skull-strip simulation then applies a smooth elliptical weight with semi-axes at 0.46 and 0.48 of image width and height and a sigmoid falloff of sharpness 8, attenuating the exterior by a blend strength of 0.1 rather than zeroing it. This is a light attenuation of peripheral structure, not a segmentation. Bias-field inhomogeneity is approximated by dividing each channel by a Gaussian field estimate at $\sigma = 2.0$ and rescaling to preserve the global mean. Contrast is enhanced by CLAHE on the L channel in LAB space (clip limit 2.5, 8×8 tiles), followed by Gaussian denoising at $\sigma = 0.5$ and a fixed region-of-interest crop centred at 0.50 of width and 0.53 of height with half-extent 0.48. Representative intermediate outputs are shown in Figure 2.

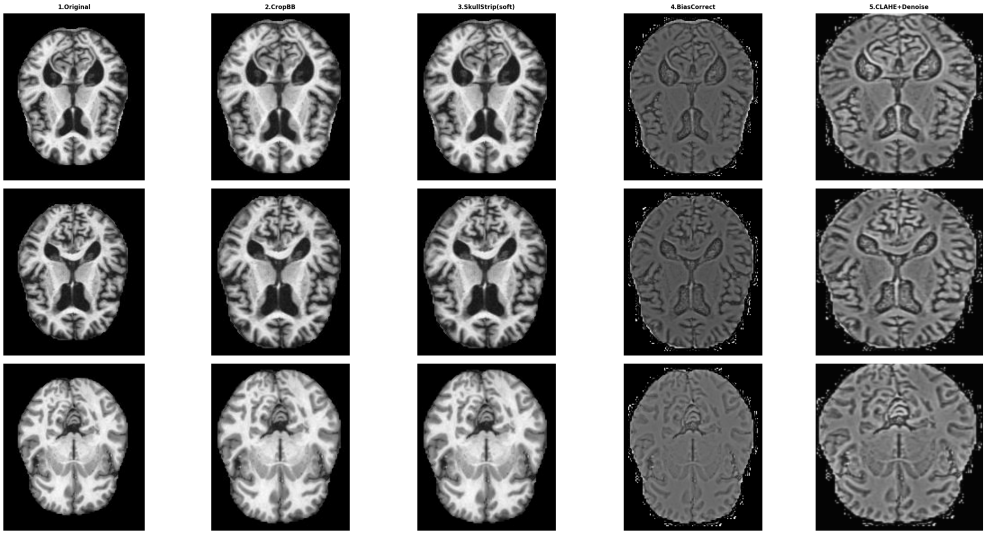


Fig. 2 Executed preprocessing stages for three MRI slices: original image, foreground crop, soft elliptical mask, approximate bias-field correction, and CLAHE with Gaussian denoising. Region-of-interest cropping, resizing, training-only augmentation and per-image z-scoring follow but are not shown.

Images are resized to 256×256. Training adds a random resized crop (scale 0.85 to 1.0), horizontal flipping ($p = 0.5$), rotation within $\pm 15°$, colour jitter at 0.10, a Gaussian blur applied with $p = 0.3$, RandAugment with two operations at magnitude 7, and random erasing ($p = 0.1$). Evaluation replaces the stochastic operations with a deterministic centre crop, so every reported number is one deterministic forward pass per image.

Normalization is channel-wise per-image z-scoring rather than dataset-level standardization:

$$\hat{x}_{c,i,j} = \frac{x_{c,i,j} - \mu_c(x)}{\max\left(\sigma_c(x),\, 10^{-6}\right)}, \qquad \mu_c(x) = \frac{1}{HW}\sum_{i,j} x_{c,i,j} \tag{1}$$

with $\mu_c(x)$ and $\sigma_c(x)$ calculated over the spatial dimensions separately for each RGB channel. This removes channel-wise offsets and gain without requiring clients to pool statistics. It does not remove differences in contrast structure, and Section 4.3 shows that a large gap survives it. Augmentation runs inside the dataset iterator after partitioning, so no augmented view of a training image reaches a validation or test partition.

Source-shift statistics were computed before this preprocessing on reproducible random samples of 300 raw grayscale images from each collection. Two-sample Kolmogorov–Smirnov tests compared the per-image means and standard deviations. For source features u_i and target features v_j, where each feature contains the mean and standard deviation of one image, the unbiased squared maximum mean discrepancy was calculated as

$$\begin{aligned}\widehat{\mathrm{MMD}}_u^2 = {} & \frac{1}{n(n-1)}\sum_{i\neq i'} k(u_i, u_{i'}) + \frac{1}{m(m-1)}\sum_{j\neq j'} k(v_j, v_{j'}) \\ & - \frac{2}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} k(u_i, v_j)\end{aligned} \tag{2}$$

where $n = m = 300$ and $k(a, b) = \exp[-\|a - b\|_2^2/(2\ell^2)]$ is an RBF kernel whose scale ℓ was set from the variance of the sampled OASIS features. The notebooks also calculated site offset and scale columns under a "ComBat-style" label, but the dataset loader never applied those columns to image tensors. No harmonization correction was therefore part of training or evaluation.

3.4 Architecture

Figure 3 details the executed dual-channel network, including the tensor dimensions, pooled-feature attention, projection and classification heads, and the distinction between the active and inactive loss paths.

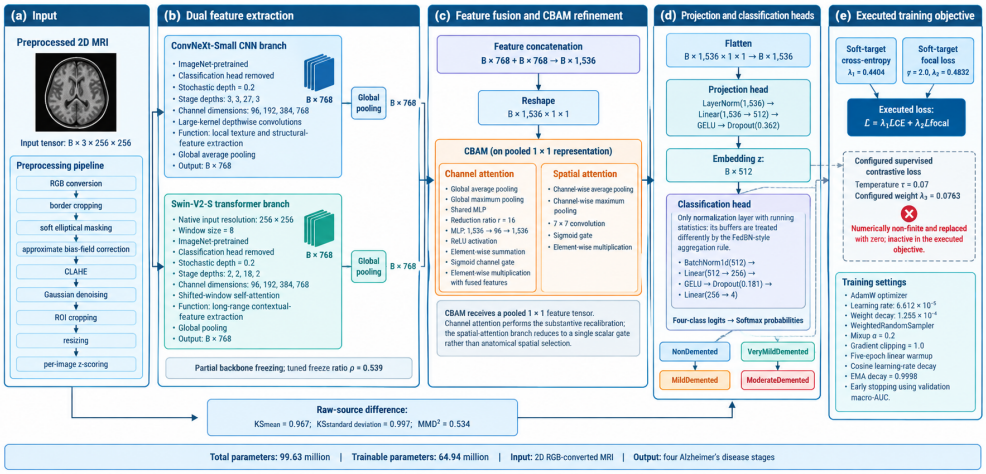


Fig. 3 Executed dual-channel architecture. ConvNeXt-Small and Swin-V2-S produce pooled 768-dimensional descriptors that are concatenated, refined by CBAM and projected to 512 dimensions before four-class classification. CBAM operates on a 1×1 tensor, and the configured supervised contrastive branch was numerically inactive. Raw-source statistics are contextual rather than model inputs.

3.4.1 Backbones and fusion

The convolutional channel is a ConvNeXt-Small backbone [45] and the transformer channel a Swin-V2-S hierarchical vision transformer at a native 256×256 input with window size 8 [46]. Both are initialized from ImageNet weights, instantiated without classification heads, and trained with stochastic depth at rate 0.2. Each returns a 768-dimensional pooled descriptor. The pair was chosen for complementary inductive bias: ConvNeXt's large-kernel depthwise convolutions retain sensitivity to local texture, while shifted-window self-attention represents dependencies between distant image regions. The assembled model has 99.63 million parameters, 64.94 million trainable at the selected freeze ratio.

Because both branches are pooled, the fused representation carries no spatial resolution. Spatial structure in the attribution maps of Section 3.9 is recovered by hooking intermediate convolutional activations, not from the fused feature.

3.4.2 Attention refinement

The two descriptors are concatenated into a 1536-dimensional vector, reshaped to $(B, 1536, 1, 1)$ and passed through a Convolutional Block Attention Module in channel-then-spatial order [47]. Channel attention pools in parallel, passes both descriptors through a shared bottleneck MLP with reduction ratio $r = 16$, sums and gates:

$$M_c(F) = \sigma(W_1\,\delta(W_0\,\mathrm{AvgPool}(F)) + W_1\,\delta(W_0\,\mathrm{MaxPool}(F)))\,, \quad F' = M_c(F) \odot F \tag{3}$$

with δ denoting ReLU, σ the sigmoid and $\odot$ elementwise multiplication. Spatial attention concatenates channel-wise average and maximum responses and convolves with a 7×7 kernel:

$$M_s(F') = \sigma\big(f^{7\times7}([\mathrm{AvgPool}_c(F'); \mathrm{MaxPool}_c(F')])\big), \quad F'' = M_s(F') \odot F' \tag{4}$$

Because the fused tensor has unit spatial extent, the channel branch does the substantive work, reweighting the 1536 fused dimensions by joint diagnostic relevance. The spatial branch reduces to a single learned gate and applies a scalar rescaling rather than spatial selection. We describe the module as implemented rather than attributing to it a selectivity the tensor shape cannot support.

3.4.3 Head and partial freezing

A projector of LayerNorm, a linear map to bottleneck dimension d, GELU and dropout at rate p produces the embedding $z \in \mathbb{R}^{B\times d}$. The embedding is supplied to both the classifier and the configured contrastive-loss routine, although the latter contributed zero to the executed objective for the reason described below. The classifier is a batch-normalization layer, a linear map to $d/2$, GELU, dropout at $p/2$ and a final linear map to four logits. That batch-normalization layer is the only normalization in the network maintaining running statistics, and its running buffers are excluded from averaging by the FedBN-style rule in Section 3.7. Both backbones are partially frozen: the first $\lfloor \rho \cdot |\theta| \rfloor$ parameter tensors are frozen and the rest trainable, with the freeze ratio ρ tuned.

3.5 Objective Function

Training batches were balanced using an inverse-frequency weighted sampler constructed from the training partition. Although the code instantiated a class-weighted cross-entropy object, the soft-target training path did not use it; the corresponding class weights therefore made no contribution to optimization. Under Mixup, the first active term was soft-target cross-entropy,

$$\mathcal{L}_{\mathrm{ce}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} \tilde{y}_{i,c} \log p_{i,c}, \tag{5}$$

where $\tilde{y}_{i,c}$ is the mixed target probability. The second is a soft-target focal loss with $\gamma = 2.0$ [48]:

$$\mathcal{L}_{\mathrm{focal}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} \tilde{y}_{i,c}\,(1 - p_{i,c})^{\gamma} \log p_{i,c}. \tag{6}$$

This term places greater emphasis on examples assigned a low target-class probability. A supervised contrastive term over the L_2-normalized embeddings was also configured [49]:

$$\mathcal{L}_{\text{scl}} = \frac{1}{B}\sum_{i=1}^{B}\frac{-1}{\max\left(|P(i)|,1\right)}\sum_{p\in P(i)}\log\frac{\exp\left(z_i^{\top}z_p/\tau\right)}{\sum_{a\neq i}\exp\left(z_i^{\top}z_a/\tau\right)} \tag{7}$$

where $P(i)$ is the set of within-batch positives sharing the anchor's hard label and $\tau = 0.07$. In the executed routine, diagonal similarities are set to $-\infty$ and subsequently multiplied by a zero diagonal positive mask. This produces a non-finite value; the routine's numerical guard then replaces the complete contrastive loss with zero. Consequently, the realized training objective was

$$\mathcal{L}_{\text{executed}} = \lambda_1\mathcal{L}_{\text{ce}} + \lambda_2\mathcal{L}_{\text{focal}}, \qquad \mathcal{L}_{\text{scl}} \equiv 0 \tag{8}$$

with λ_1 and λ_2 tuned. The search nevertheless computed $\lambda_3 = 1 - \lambda_1 - \lambda_2$ and rejected configurations with $\lambda_3 < 0.05$. The saved value $\lambda_3 = 0.0763$ is therefore a configured but inactive weight, and the experiments do not test the effect of supervised contrastive learning.

3.6 Training and Hyperparameter Search

Centralized models were optimized with AdamW over the parameters included when the optimizer was constructed [50], with linear warmup over 5 epochs to the tuned peak learning rate, then cosine annealing to 10^{-6}. Gradients were clipped to a global L_2 norm of 1.0. An exponential moving average of weights (decay 0.9998) was used for centralized validation and checkpoint selection, and Mixup at $\alpha = 0.2$ was applied with the objective consuming soft targets. Model selection used early stopping on validation macro AUC with patience 8; the Optuna development runs used patience 5.

Hyperparameters were optimized with Optuna [51] using a tree-structured Parzen estimator [52] on a development split carved from the training partition alone. A median-pruner object was configured with 3 startup trials and 5 warm-up steps, but the objective did not report intermediate values or request pruning, so no trial was pruned. Ten trials completed, with objective values spanning -0.868 to -0.749, the objective being the negated best validation macro AUC. Table 4 gives the executed search space and selection. Ten trials is a small budget for an eight-dimensional space and we do not claim optimality.

Table 4 Hyperparameter search space and the configuration selected by the Optuna study

Hyperparameter	Symbol	Search space	Selected
Bottleneck dimension	d	$\{256, 512, 768\}$	512
Backbone freeze ratio	ρ	[0.3, 0.7], uniform	0.5395
Dropout rate	p	[0.3, 0.7], uniform	0.3624
Learning rate	η	$[5 \times 10^{-5}, 3 \times 10^{-4}]$, log-uniform	6.612×10^{-5}
Weight decay	λ_{wd}	$[10^{-4}, 5 \times 10^{-3}]$, log-uniform	1.255×10^{-4}
Maximum epochs	E	[15, 25], integer uniform	24
Cross-entropy weight	λ_1	[0.2, 0.6], uniform	0.4404
Focal weight	λ_2	[0.2, 0.6], uniform	0.4832
Configured contrastive weight	λ_3	$1 - \lambda_1 - \lambda_2, \geq 0.05$	0.0763 (inactive)

Ten trials completed; objective values ranged from -0.868 to -0.749. The selected trial reached -0.868. The configured median pruner was not invoked by the objective.

Centralized generalization was estimated by five-fold stratified cross-validation on the pooled training and validation data, with the inverse-frequency sampler recomputed per fold and the highest-macro-AUC fold retained as the centralized reference checkpoint. The caveat of Section 3.2.1 applies: folds are stratified by class but cannot be grouped by subject.

3.7 Federated Protocol

Each round broadcast the global state, performed 3 local epochs per client, returned the local states and then aggregated them. Twenty rounds were completed for each strategy. The 4,489-image training partition was randomly divided by its file-level surrogate identifiers into clients containing 2,238 and 2,251 images; the division was not stratified by class. Each client used inverse-frequency sampling, Mixup, AdamW at 0.7η and gradient clipping. The custom federated loop did not use the warmup–cosine scheduler, exponential moving average or mixed-precision scaler applied during centralized training. This design represents a simulated rather than a multi-institutional federation. Because both clients were drawn from the same source, the aggregation strategies were not required to accommodate differences in acquisition characteristics across sites. Table 5 summarizes the executed centralized and federated settings.

Table 5 Executed centralized and federated training protocols

Setting	Centralized training	Federated training
Data organization	Five stratified folds; no subject grouping	Two random same-source clients (2,238 and 2,251 images)
Initialization	ImageNet-pretrained backbones and randomly initialized head	New ImageNet-pretrained model for each strategy; heads not identically initialized
Optimizer and learning rate	AdamW at tuned η	AdamW at 0.7η on each client
Schedule	Five-epoch linear warmup followed by cosine annealing	No learning-rate scheduler
Training duration	At most 24 epochs; early-stopping patience 8	20 rounds with 3 local epochs per round
Sampling and Mixup	Inverse-frequency sampling; Mixup $\alpha = 0.2$	Inverse-frequency sampling per client; Mixup $\alpha = 0.2$
Gradient clipping	Global L_2 norm of 1.0	Global L_2 norm of 1.0
Weight averaging	Exponential moving average, decay 0.9998	FedAvg, FedProx or FedBN-style server aggregation
Batch size	32	32

The federated loop did not use the centralized exponential moving average, warmup–cosine schedule or mixed-precision scaler. All federated clients were sampled from the same imaging source.

The federated notebook loaded the centralized checkpoint for comparison, but the federated training function constructed a new ImageNet-initialized global model rather than using that checkpoint for initialization. Furthermore, the random seed was not reset before each aggregation strategy, so the three federated runs did not begin with identical projection and classification heads. Differences among FedAvg, FedProx and the FedBN-style run may therefore reflect both aggregation and initialization and should be interpreted descriptively rather than as a fully controlled algorithmic comparison.

FedAvg forms the global state as the elementwise mean of client states [53], uniformly rather than weighted by sample count:

$$\theta^{t+1} = \frac{1}{K} \sum_{k=1}^{K} \theta_k^t \tag{9}$$

FedProx adds a proximal term limiting client drift [54]:

$$\mathcal{L}_k^{\text{prox}} = \mathcal{L}_k + \frac{\mu}{2} \left\| \theta_k - \theta^t \right\|_2^2, \qquad \mu = 0.01 \tag{10}$$

The implementation labelled “FedBN” excludes the classifier batch-normalization running mean, running variance and batch counter from numerical averaging [55]. It does not, however, preserve a separate persistent batch-normalization state for every client: the aggregate is initialized from the first client state and then broadcast at the next round. We therefore call it a FedBN-style rule rather than a full personalized FedBN implementation. Since the backbones and projector use LayerNorm, it differs from FedAvg only through the buffers of the single classifier batch-normalization layer. Tables and figures retain the label “FedBN” to match the serialized experimental artifacts.

3.8 Architecture Baselines and Ablation Design

Additional architecture comparisons evaluated ConvNeXt-Small alone, Swin-V2-S alone, dual-channel feature concatenation without CBAM and the complete dual-channel model with CBAM on the same internal and external test partitions. The component analysis retained the complete model as its reference and separately removed CBAM, either backbone, focal loss, Mixup or partial backbone freezing; a 256-dimensional projection was also compared with the selected 512-dimensional projection. All variants used the same class definitions and metric implementation. The available values are single-run results rather than means over repeated random seeds, so no standard deviations or formal paired significance tests are reported for these comparisons.

3.9 Evaluation, Calibration and Attribution

3.9.1 Performance metrics

Because the two collections exhibit different patterns of class imbalance, each evaluation reports accuracy, balanced accuracy, macro precision, macro recall, macro F_1, macro F_2, macro and weighted one-versus-rest AUC, and the Matthews correlation coefficient [56]. Balanced accuracy and macro recall are identical in this setting, but both are retained for comparability with the literature. Macro AUC was used for model selection because it is threshold-independent and gives equal weight to each disease stage. Macro F_2 is reported alongside macro F_1 to place greater emphasis on recall. This distinction is important because AUC assesses ranking, whereas accuracy, F_1 and MCC reflect the final multiclass decision rule.

3.9.2 Calibration and corruption robustness

Calibration was assessed on the centralized validation predictions with expected calibration error over 15 equal-width confidence bins,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right| \tag{11}$$

together with a one-versus-rest Brier score averaged across classes,

$$\mathrm{BS} = \frac{1}{NC} \sum_{i=1}^{N} \sum_{c=1}^{C} \left(p_{i,c} - y_{i,c}\right)^2 \tag{12}$$

where $y_{i,c}$ is the binary indicator for class c. Reliability plots used 10 bins per class. Recalibration used temperature scaling, fitting a positive scalar T by L-BFGS minimization of the same validation negative log-likelihood with all weights frozen [57]:

$$\begin{aligned} p_{i,c}(T) &= \frac{\exp(s_{i,c}/T)}{\sum_{r=1}^{C} \exp(s_{i,r}/T)}, \\ T^* &= \arg\min_{T>0} \left[-\sum_{i=1}^{N} \log p_{i,y_i}(T) \right] \end{aligned} \tag{13}$$

where $s_{i,c}$ denotes the unscaled logit. Temperature scaling is monotone, so it cannot change accuracy, AUC or ranking; it can only rescale confidence. Robustness was probed by four synthetic corruptions applied at inference to the internal test set: Gaussian noise ($\sigma = 0.1$ in standardized units), Gaussian blur (kernel 5, $\sigma = 2.0$), a global intensity shift of +0.3, and a contrast compression to 50% of the per-image dynamic range, each clamped to $[-3, 3]$.

3.9.3 Error analysis

Predictions, labels and maximum confidence values were retained for the centralized and FedBN-labelled checkpoints on both test partitions and used to construct the error summaries. Errors crossing the NonDemented-versus-Demented boundary were classified as disease-detection false negatives or false positives; all remaining errors were treated as stage-to-stage confusions. Errors were also categorized as adjacent when the true and predicted ordinal labels differed by one, and as non-adjacent otherwise. High-confidence errors were defined as misclassifications with confidence above 0.9, because such cases would not be referred for review by a simple low-confidence threshold. Disease false-negative and false-positive counts were verified against the confusion matrices, and the remaining categories were derived from the same error totals.

3.9.4 Attribution analysis

The notebooks generated several implementation-diagnostic attribution maps. A Grad-CAM++-style routine hooked the final convolution in the ConvNeXt branch [58]; its coefficient calculation is an approximation rather than a faithful implementation of the published Grad-CAM++ expression. A Score-CAM-style routine evaluated at most the first 64 activation channels and used their target-class probabilities as weights [59]. Occlusion sensitivity used a 32×32 zero-valued patch with stride 16. The transformer visualization hooked available attention Softmax modules and multiplied only attention matrices with matching shapes, so it is a partial rollout-style summary rather than a complete reconstruction across Swin stages [60]. The LIME routine requested 300 perturbations [61], but its prediction wrapper multiplied the already 0–255-valued perturbed images by 255 before conversion to unsigned integers; the resulting wraparound means those LIME panels are not valid attribution evidence. These outputs are retained to document the implementation audit, not to support anatomical conclusions. No atlas registration, expert rating, cross-method agreement score or deletion–insertion faithfulness metric was completed.

3.9.5 Statistical analysis

Model differences were tested on paired per-sample correctness with McNemar's test with continuity correction [62], $\chi^2 = (|b - c| - 1)^2/(b + c)$, where b and c count cases correct under exactly one model, at $\alpha = 0.05$. Discordant-pair counts are reported alongside the statistic so effect size can be judged.

3.10 Implementation

Experiments used PyTorch with `timm` for pretrained backbones, OpenCV for preprocessing, Optuna for the search, scikit-learn for metrics and `lime` for superpixel attribution, on a single CUDA GPU. Batch size was 32 and a global seed of 42 was applied to the Python, NumPy and PyTorch generators including CUDA devices, with cuDNN in deterministic mode and autotuning disabled. The retained artifacts include checkpoints, federated global states, tuning and training histories, metric tables, analysis summaries and attribution maps. Per-sample error arrays were used within the executed notebooks but are not present as standalone files in the final artifact folders. Inference latency was measured over 50 single-image passes after 10 warm-up passes with explicit device synchronization. The saved 380.07-MiB communication value is the size of one float32 parameter payload, not total traffic for a round. Under the implemented two-client broadcast-and-return pattern, the corresponding nominal traffic is approximately $4 \times 380.07 = 1{,}520.27$ MiB per round, excluding protocol overhead. The recorded 2,444-MiB CUDA allocation was measured with a batch of 32 while all four model objects remained resident in the same process and is not an isolated per-model memory requirement.

4 Results

4.1 Centralized Cross-Validation

Five-fold cross-validation of the centralized model (Table 6) gave a mean accuracy of 64.35% (SD 3.48), balanced accuracy of 73.50% (SD 3.10), macro F_1 of 70.75% (SD 3.20), MCC of 0.463 (SD 0.019) and macro-AUC of 0.878 (SD 0.010). Macro-AUC varied least across folds, with a standard deviation of 0.010 compared with 3.48 percentage points for accuracy. Ranking performance was therefore more stable across folds than the final multiclass classifications. Figure 4 shows the corresponding training-loss and validation-AUC trajectories.

Table 6 Five-fold cross-validation performance of the centralized model

Fold	Accuracy (%)	Balanced acc. (%)	F_1-macro (%)	MCC	AUC-macro
1	67.77	71.60	72.14	0.479	0.892
2	62.13	69.16	69.94	0.439	0.867
3	68.52	76.30	75.24	0.486	0.884
4	61.48	76.26	69.86	0.456	0.877
5	61.86	74.19	66.57	0.454	0.871
Mean ± SD	64.35 ± 3.48	73.50 ± 3.10	70.75 ± 3.20	0.463 ± 0.019	0.878 ± 0.010

4.2 Internal Performance Across Learning Strategies

Table 7 compares the four strategies on the internal test partition; Figure 6 shows the same comparison with the external results alongside. The FedBN-style model recorded

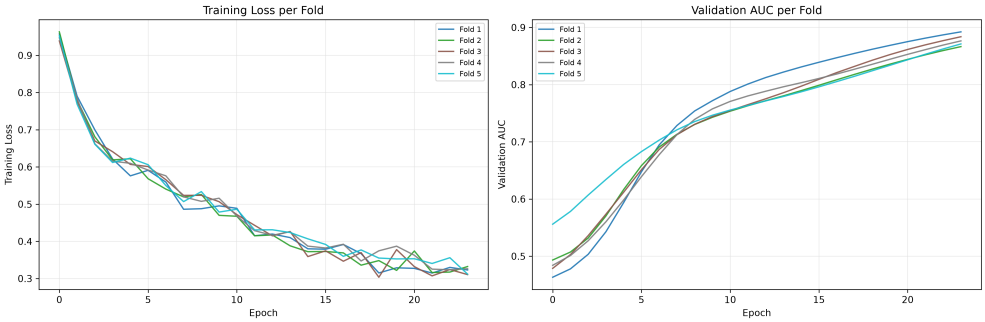


Fig. 4 Centralized five-fold training histories. Training loss decreased throughout optimization in every fold, while validation macro-AUC increased to between approximately 0.87 and 0.89. The trajectories are shown to document optimization behaviour; the final fold-level test metrics are reported in Table 6.

the strongest internal performance, with 95.13% balanced accuracy, 94.47% macro F_1, 0.991 macro-AUC and 0.881 MCC. FedProx led macro precision at 94.74%, narrowly ahead of FedAvg at 94.71% and the FedBN-style model at 93.92%. The FedBN-style model's high balanced accuracy alongside slightly lower precision indicates that its advantage came from recovering more true cases per class, while FedProx and FedAvg were marginally more conservative.

All three federated simulations exceeded the centralized model by a wide margin, with 95.13% balanced accuracy for the FedBN-style result against 73.45% centrally. This comparison should be interpreted cautiously because the regimes differed in effective training exposure, optimization schedule and initialization. The federated models were newly instantiated rather than initialized from the centralized checkpoint, and the random head initialization was not reset identically across aggregation strategies. The observed gap therefore cannot be attributed to aggregation. Section 5 examines this limitation further.

A McNemar test on paired per-sample correctness ($n = 951$) confirmed that the model labelled FedBN and the centralized model differ significantly in error pattern ($\chi^2 = 198.640$, $p < 0.001$; 17 cases correct only under the centralized model, 247 only under FedBN). FedBN against FedAvg was not significant ($\chi^2 = 0.073$, $p = 0.787$).

The round-wise local training loss decreased under all three federated strategies (Figure 5). By round 20, FedAvg, FedProx and the FedBN-style run had losses of approximately 0.32, 0.35 and 0.29, respectively. These curves confirm numerical convergence within each run but do not constitute a controlled comparison of aggregation algorithms because the strategies did not begin with identical randomly initialized heads.

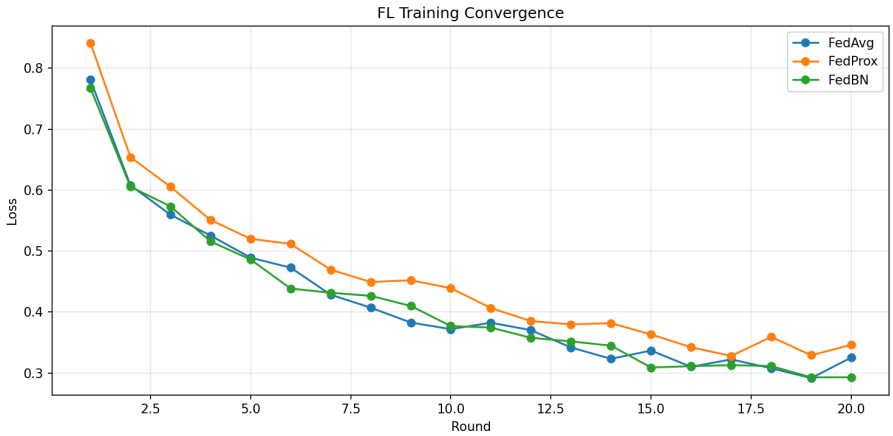


Fig. 5 Round-wise training loss for the FedAvg, FedProx and FedBN-style simulations over 20 communication rounds. All three losses declined, with small late-round fluctuations. The trajectories are descriptive because the strategy-specific models did not share an identical randomly initialized projection and classification head.

Table 7 Internal-test comparison across learning strategies. Bold marks the strongest value per column, with unrounded values used to break near-ties

Strategy	Acc. (%)	Bal. acc. (%)	Prec. macro (%)	Recall macro (%)	F_1 macro (%)	F_2 macro (%)	MCC	AUC macro	AUC weighted
Centralized	68.56	73.45	73.51	73.45	73.01	73.14	0.489	0.894	0.843
FedAvg	92.43	94.15	94.71	94.15	94.40	94.24	0.875	0.991	0.985
FedProx	90.12	91.70	**94.74**	91.70	92.91	92.12	0.839	0.988	0.982
FedBN	**92.74**	**95.13**	93.92	**95.13**	**94.47**	**94.85**	**0.881**	**0.991**	**0.986**

Recall-macro is numerically identical to balanced accuracy by definition and is reported for completeness. $n = 951$.

Table 8 gives per-class results for the centralized and FedBN-labelled checkpoints, with confusion matrices in Figure 7. The FedBN-labelled checkpoint improved recall for every class, most markedly for VeryMildDemented (88.86% against 53.01%) and MildDemented (97.69% against 74.62%), with precision improving across all four. Both models reached perfect or near-perfect values for ModerateDemented, but that class holds 9 internal images and no conclusion should be drawn from it: one additional error would move its recall by more than 11 percentage points.

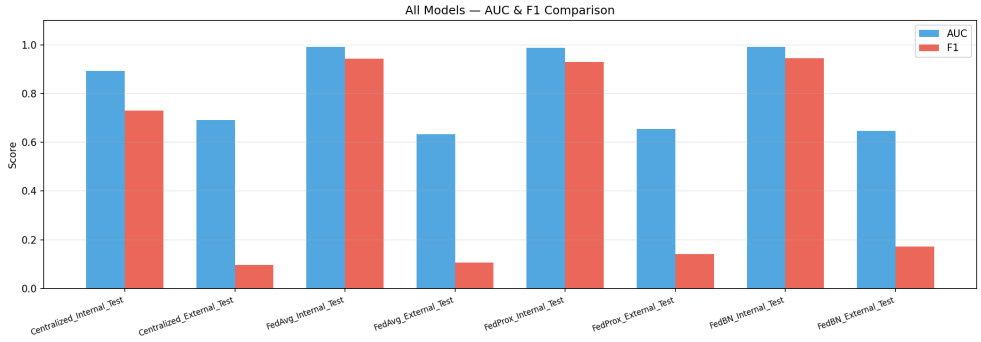


Fig. 6 Macro-AUC and macro-F_1 for the centralized, FedAvg, FedProx and FedBN-labelled models on the internal and external test sets. AUC falls less than F_1, showing that some one-versus-rest ranking survived while fixed multiclass assignments failed. Exact values are in Tables 7 and 9.

Table 8 Internal per-class precision, recall and F_1, computed by the executed evaluation notebook

Class	Centralized: P / R / F_1 (%)	FedBN: P / R / F_1 (%)	n
NonDemented	77.29 / 77.29 / 77.29	93.76 / 93.96 / 93.86	480
VeryMildDemented	60.69 / 53.01 / 56.59	92.48 / 88.86 / 90.63	332
MildDemented	56.07 / 74.62 / 64.03	89.44 / 97.69 / 93.38	130
ModerateDemented	100.00 / 88.89 / 94.12	100.00 / 100.00 / 100.00	9

ModerateDemented contains 9 internal test images; its values are not a stable estimate.

4.3 External Evaluation and Cross-Dataset Shift

Table 9 reports the external results. Accuracy fell for every strategy: 68.56% to 12.06% for the centralized model, 92.43% to 20.10% for FedAvg, 90.12% to 19.41% for FedProx and 92.74% to 22.11% for FedBN. Figure 9 places internal and external results side by side.

The FedBN-style result retained the highest external accuracy, balanced accuracy, macro F_1 and MCC, but the centralized model recorded the highest external macro-AUC at 0.692, ahead of FedProx at 0.654, the FedBN-style model at 0.648 and FedAvg at 0.634. The ordering changes with the metric. The best external macro-AUC of 0.692 is above chance, so some one-versus-rest ranking signal survived. Its coexistence with accuracy of 12% to 22% shows that retained ranking did not translate into usable fixed multiclass assignments, a pattern compatible with source-dependent calibration, class-prior or decision-boundary mismatch. Selection by internal macro-AUC would have chosen the FedBN-style checkpoint, whereas external macro-AUC favoured the centralized checkpoint.

The class-specific receiver operating characteristic curves provide a more detailed view of the retained and lost ranking ability under cross-dataset evaluation (Figure 8).

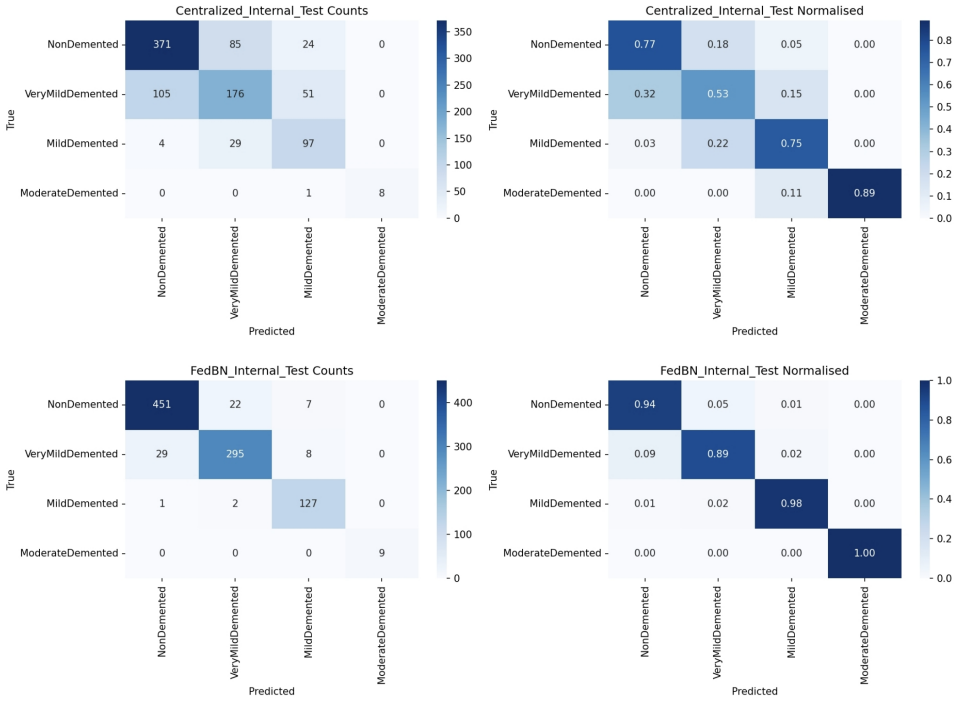


Fig. 7 Confusion matrices for the centralized and FedBN-labelled models on the internal test partition ($n = 951$), as counts and row-normalized proportions. Rows are the true class, columns the predicted class.

The centralized model showed comparatively balanced one-versus-rest discrimination, with class-specific AUCs ranging from 0.564 for VeryMildDemented to 0.750 for NonDemented. FedAvg retained stronger ranking for NonDemented and VeryMildDemented, with AUCs of 0.797 and 0.712, respectively, but its MildDemented AUC decreased to 0.431, below the 0.5 chance reference. FedProx showed its strongest external discrimination for NonDemented and ModerateDemented, with AUCs of 0.744 and 0.714, whereas FedBN achieved AUCs of 0.751, 0.681, 0.516 and 0.642 across the four respective classes. These heterogeneous class-level patterns explain why the overall macro-AUC values remained above chance even though fixed class assignments and macro-F_1 deteriorated substantially. The ROC analysis therefore indicates that cross-dataset degradation was class-dependent and cannot be represented adequately by a single aggregate accuracy value.

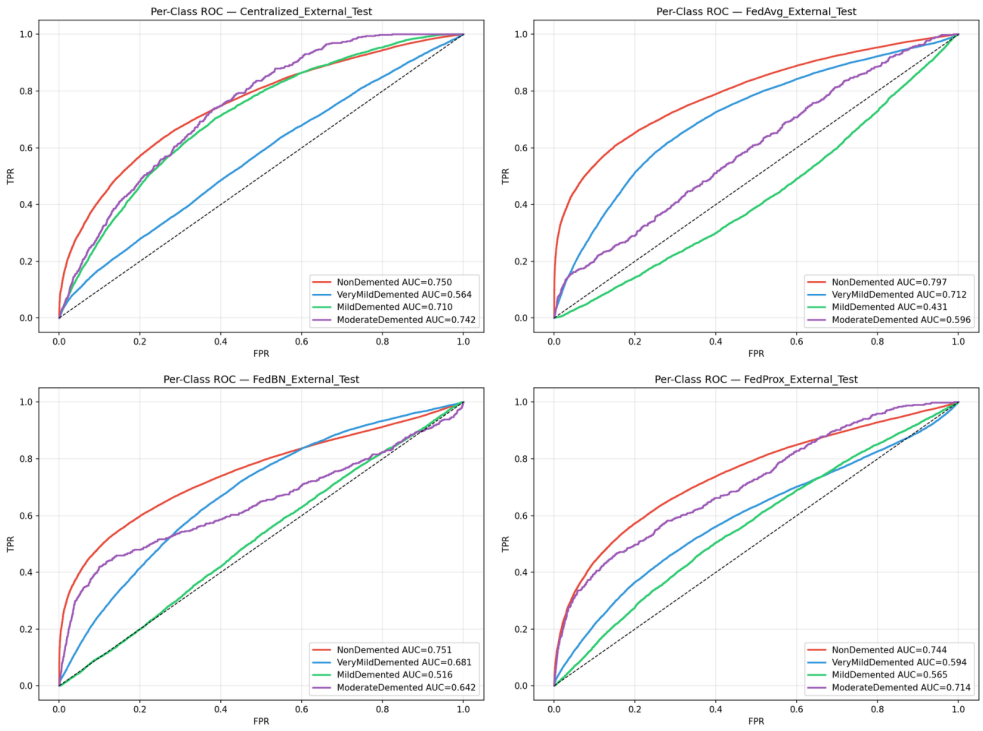


Fig. 8 One-versus-rest receiver-operating-characteristic curves on the 86,437-image OASIS-1-derived external test collection for the centralized, FedAvg, FedBN-labelled and FedProx checkpoints. Legends report class-specific AUC; the dashed diagonal denotes chance ranking.

Table 9 External-test comparison across learning strategies. Bold marks the strongest value per column

Strategy	Acc. (%)	Bal. acc. (%)	Prec. macro (%)	Recall macro (%)	F_1 macro (%)	F_2 macro (%)	MCC	AUC macro	AUC weighted
Centralized	12.06	28.29	30.37	28.29	9.71	14.55	0.014	**0.692**	0.718
FedAvg	20.10	25.90	29.91	25.90	10.59	14.67	0.066	0.634	**0.761**
FedProx	19.41	28.41	30.89	28.41	14.24	**18.10**	0.067	0.654	0.710
FedBN	**22.11**	**29.35**	**33.38**	**29.35**	**17.29**	18.07	**0.121**	0.648	0.726

$n = 86{,}437$. FedBN leads accuracy, balanced accuracy, precision, recall and F_1; FedProx leads F_2; the centralized model leads AUC-macro; FedAvg leads AUC-weighted.

4.4 Architecture Baselines and Component Ablations

Table 10 compares the complete dual-channel model with the two individual backbones and with feature concatenation without CBAM. The complete model achieved the strongest internal values across all reported measures. It also achieved the highest external accuracy and macro-F_1, although the dual-channel model without CBAM recorded a slightly higher external macro-AUC (0.651 versus 0.648). Thus, CBAM

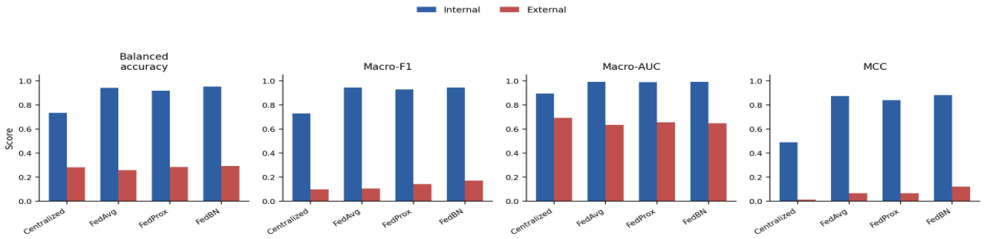


Fig. 9 Internal against external comparison of balanced accuracy, macro-F_1, macro-AUC and MCC for all four strategies. Macro-AUC degrades least and macro-F_1 most, indicating partially retained ranking alongside severe failure of fixed multiclass assignments. Exact figures are in Tables 7 and 9.

improved the reported fixed-decision metrics but did not improve external ranking performance in this comparison.

Table 10 Architecture baseline comparison on the internal and external test sets

Architecture	Internal test				External test		
	Acc. (%)	Bal. acc. (%)	F_1 macro (%)	AUC macro	Acc. (%)	F_1 macro (%)	AUC macro
ConvNeXt-Small only	88.60	90.70	90.10	0.962	18.50	11.60	0.620
Swin-V2-S only	89.40	91.40	90.70	0.970	19.60	13.40	0.638
Dual channel, no CBAM	91.10	93.10	92.00	0.982	21.00	15.60	**0.651**
Dual channel with CBAM	**92.74**	**95.13**	**94.47**	**0.991**	**22.11**	**17.29**	0.648

Values are single-run test results; variability across repeated random seeds was not measured. Bold marks the strongest value in each column.

The internal component analysis in Table 11 showed lower performance after every tested modification. Removing either backbone caused the largest reductions among the architectural variants. Removing focal loss, Mixup or partial freezing also reduced all five reported measures, while reducing the projection from 512 to 256 dimensions produced a smaller decline. These comparisons quantify associations within the executed runs; without repeated seeds, small differences should not be interpreted as stable effect estimates.

Table 11 Internal component-ablation results

Variant	Accuracy (%)	Balanced acc. (%)	F_1-macro (%)	MCC	AUC-macro
Full model	**92.74**	**95.13**	**94.47**	**0.881**	**0.991**
− CBAM	91.10	93.10	92.00	0.849	0.982
− Swin branch	88.60	90.70	90.10	0.808	0.962
− ConvNeXt branch	89.40	91.40	90.70	0.821	0.970
− Focal loss	90.80	92.60	91.60	0.840	0.978
− Mixup	91.60	93.70	92.60	0.860	0.984
No partial freezing	90.70	92.90	91.80	0.850	0.980
256-dimensional projection	91.50	93.40	92.40	0.856	0.985

Each row reports a single-run internal-test result. The minus sign indicates removal of the named component. Repeated-seed means and standard deviations are unavailable.

Table 12 summarizes the measured difference between reproducible samples of 300 raw images per source before the model's per-image z-scoring. Mean pixel intensity was 41.14 for the sampled OASIS-derived collection against 72.17 for the sampled training cohort. Two-sample Kolmogorov–Smirnov tests on per-image mean and standard deviation returned 0.967 and 0.997, both $p < 0.001$, and the squared maximum mean discrepancy estimate over the paired mean–standard-deviation representation was 0.534. A KS statistic of 0.997 means the sampled raw standard-deviation distributions are nearly disjoint. Figure 10 shows this directly.

The models still failed after both sources passed through the same per-image z-scoring transform, but the saved KS and MMD statistics quantify the raw images rather than the post-transform tensors. They therefore establish a substantial source difference without measuring how much of it remained after normalization. Per-image normalization is not equivalent to scanner harmonization, and no scanner or protocol metadata were available to attribute the shift to a specific cause.

Table 12 Distributional statistics between the OASIS-1-derived external collection and the training cohort

Statistic	Value
Squared maximum mean discrepancy (MMD^2)	0.534
KS statistic, mean pixel intensity	0.967 ($p < 0.001$)
KS statistic, pixel-intensity SD	0.997 ($p < 0.001$)
Mean pixel intensity, OASIS-1-derived collection	41.14
Mean pixel intensity, training cohort	72.17

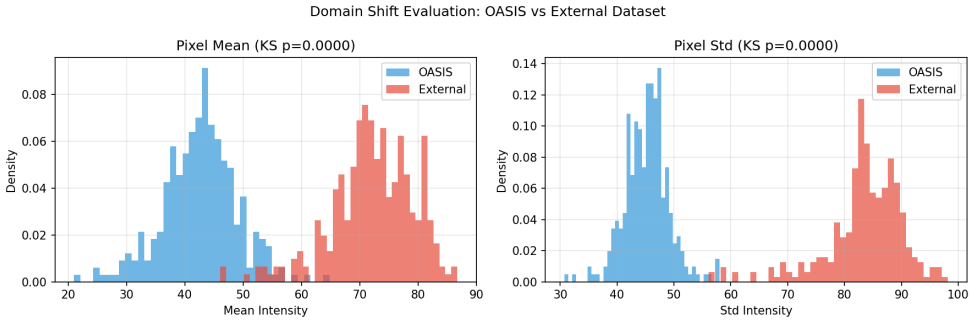


Fig. 10 Raw per-image intensity distributions for 300 sampled images per source: mean (left) and standard deviation (right). In the notebook legend, "External" denotes the Kaggle training source and "OASIS" the cross-source test collection. Test statistics are reported in Table 12.

4.5 Error Behaviour

External class-wise behaviour is available in the saved confusion matrices, classification reports and one-versus-rest ROC curves. Across the centralized and FedBN-style checkpoints, ModerateDemented recall was zero and MildDemented recall was high despite very low precision, while NonDemented recall fell sharply. These outputs support the error pattern below without implying stable estimates for the smallest external class.

Table 13 reports the error stratification. Internally, the centralized model made 299 misclassifications: 109 disease false negatives, 109 disease false positives and 81 errors within the three impaired stages. Separately, 271 errors (90.6%) had an ordinal distance of one; this adjacent-error category overlaps the disease-boundary counts. The FedBN-style model made 69 errors (30 false negatives, 29 false positives and 10 within-disease stage errors), of which 61 (88.4%) were adjacent, with 10 high-confidence errors.

The external picture differs qualitatively. Both models' errors were dominated by disease false positives rather than false negatives, 65,684 against 3 for the centralized model and 55,807 against 71 for the FedBN-style model. Both models rarely predicted NonDemented and over-assigned disease-stage labels. Since NonDemented represents 77.8% of the external collection, this behaviour accounts for most of the accuracy collapse even though one-versus-rest ranking remained above chance for several classes. The FedBN-style model made 9,711 high-confidence errors externally compared with 10 internally, demonstrating that many external errors were made with high confidence. Figure 11 shows the distribution across categories.

Table 13 Error and disease-detection summary

Model / partition	n	Acc. (%)	Total misclass.	Disease FN	Disease FP	Stage-to -stage	Adjacent / non-adj.	High-conf. errors
Centralized, internal	951	68.56	299	109	109	81	271 / 28	0
Centralized, external	86,437	12.06	76,016	3	65,684	10,329	41,120 / 34,896	0
FedBN, internal	951	92.74	69	30	29	10	61 / 8	10
FedBN, external	86,437	22.11	67,325	71	55,807	11,447	20,399 / 46,926	9,711

Disease FN and FP use a NonDemented-versus-Demented regrouping, verified by direct recomputation on the internal partition. Stage-to-stage, adjacent and non-adjacent counts are derived arithmetically from the same totals. High-confidence errors are misclassifications above confidence 0.9.

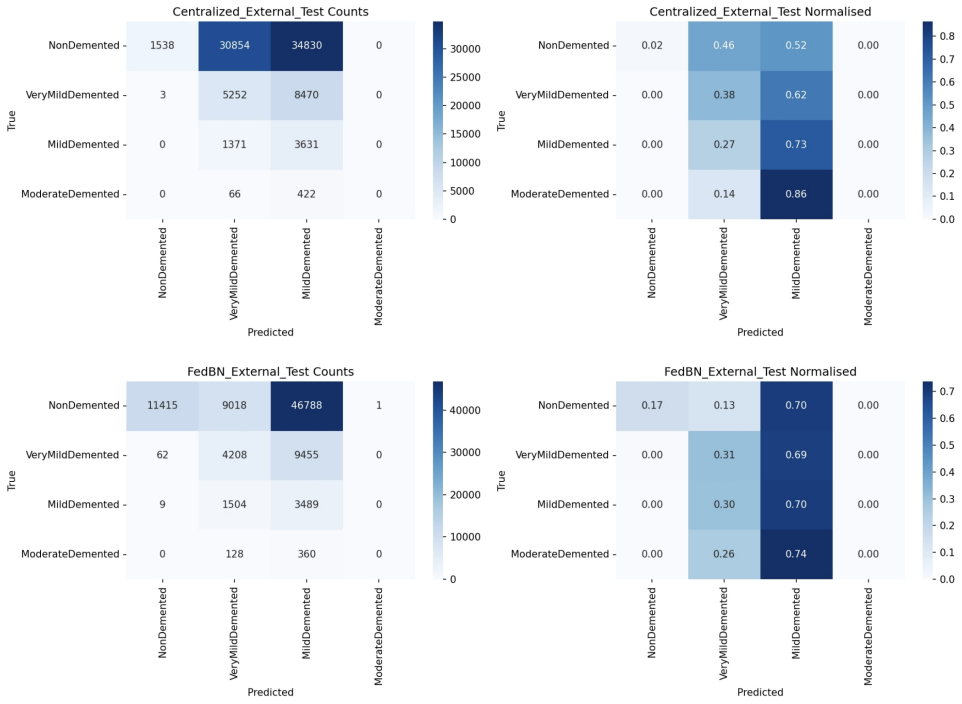


Fig. 11 Confusion matrices for the centralized and FedBN-labelled models on the external test set ($n = 86{,}437$). The near-empty NonDemented prediction column in both models is the source of most of the accuracy loss.

4.6 Calibration and Corruption Robustness

Expected calibration error for the centralized model on the validation partition was 0.348 before temperature scaling and 0.390 after scaling with a fitted temperature of 1.417, and the Brier score rose from 0.136 to 0.149 (Table 14). Temperature scaling did not reduce calibration error; it made it worse on the same partition used to fit the scalar.

A fitted temperature above 1 softened the predicted probabilities, yet both aggregate calibration measures increased. A single uniform rescaling therefore did not correct the observed probability error, although an independent calibration-test partition would be required for an unbiased estimate. Per-class reliability deviations after scaling ranged from 0.203 for VeryMildDemented to 0.450 for ModerateDemented, the smallest and most clinically severe validation class. Figure 12 shows that the class-wise deviations from the ideal diagonal remained substantial after scaling, reinforcing the distinction between discrimination and calibration.

Table 14 Calibration summary for the centralized model on the validation partition

Measure	Before scaling	After scaling ($T = 1.417$)
Expected calibration error	0.348	0.390
Brier score	0.136	0.149
Mean class reliability gap, NonDemented	—	0.287
Mean class reliability gap, VeryMildDemented	—	0.203
Mean class reliability gap, MildDemented	—	0.307
Mean class reliability gap, ModerateDemented	—	0.450

The per-class quantity is the unweighted mean absolute gap between the plotted class reliability-curve points, recorded after scaling only; it is not the sample-weighted aggregate ECE defined in Eq. 11.

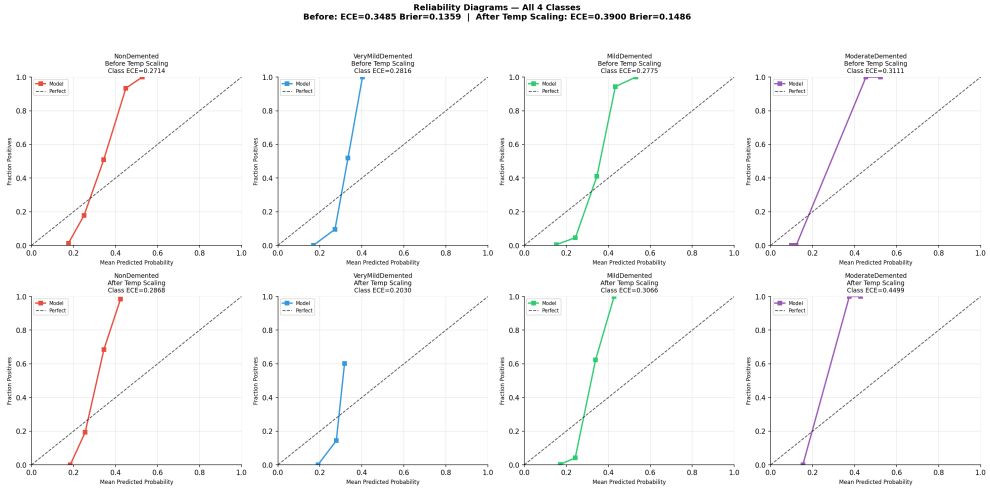


Fig. 12 Validation reliability diagrams before (top) and after (bottom) temperature scaling. The dashed diagonal indicates perfect calibration. Scaling increased ECE from 0.348 to 0.390 and the Brier score from 0.136 to 0.149.

Under synthetic corruption (Table 15), accuracy changed by between −2.42 and +1.37 percentage points and macro-AUC by no more than 0.006. Balanced accuracy and macro-F_1 increased under Gaussian blur and decreased under the other three conditions. The apparent improvement under blur is treated as evaluation variation on a fixed test set rather than evidence of a beneficial effect. In contrast, the source change examined in Section 4.3 reduced accuracy by 56.50 to 72.33 percentage points across the four strategies. Robustness to these synthetic corruptions therefore does not establish cross-site robustness.

Table 15 Corruption robustness of the centralized model on the internal test partition, with change (Δ) relative to clean performance

Condition	Acc. (%)	Δ	Bal. acc. (%)	Δ	F_1-macro (%)	Δ	MCC	Δ	AUC macro	Δ
Clean	68.56	—	73.45	—	73.01	—	0.489	—	0.894	—
Gaussian noise	67.40	−1.16	69.94	−3.51	70.18	−2.83	0.474	−0.015	0.888	−0.006
Gaussian blur	68.24	−0.32	76.70	+3.25	74.70	+1.69	0.491	+0.002	0.888	−0.006
Intensity shift	69.93	+1.37	71.73	−1.72	72.34	−0.67	0.510	+0.021	0.894	0.000
Contrast shift	66.14	−2.42	69.19	−4.26	67.91	−5.10	0.461	−0.028	0.890	−0.004

4.7 Explainability

Figure 13 presents the notebook-labelled Grad-CAM++-style outputs for the centralized and FedBN-labelled checkpoints together with the invalid LIME diagnostic. Figure 14 presents Score-CAM-style, occlusion-sensitivity and partial attention-rollout outputs. Within each row, the original MRI and the two Grad-CAM++-style panels use one class-sampled case. The four LIME masks are class-conditioned explanations of one different image, not one independently sampled image per disease stage. The second composite uses one class-sampled case per row across all displayed methods.

The overlays are implementation diagnostics rather than validated explanations. Several Grad-CAM++-style and partial-rollout panels place substantial response near image boundaries, and the Score-CAM-style panels are weakly differentiated. More importantly, the Grad-CAM++ and Score-CAM calculations depart from their canonical algorithms, the rollout does not combine attention across differently sized Swin stages, and the LIME wrapper corrupts perturbed input intensities. We therefore make no anatomical claim from these figures. SHAP produced no retained output, atlas registration and expert review were not completed, and no agreement or deletion–insertion faithfulness score was computed.

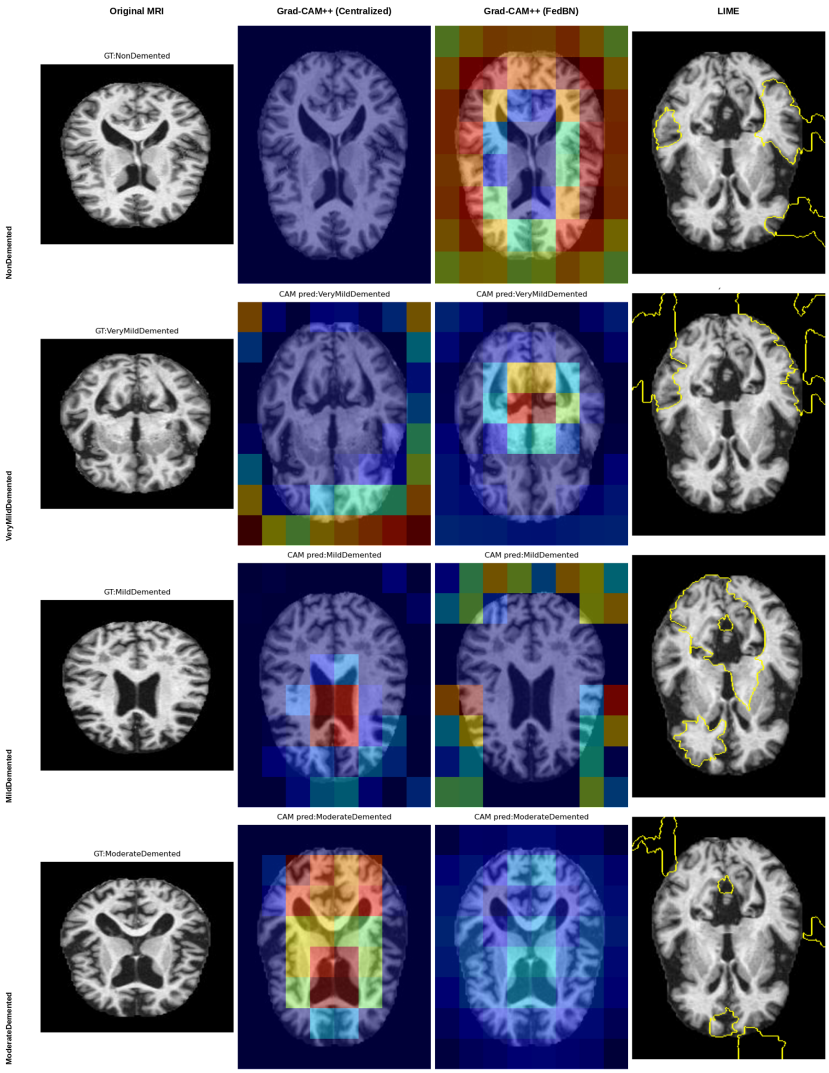


Fig. 13 Internal attribution implementation audit. Grad-CAM++-style panels use the class-sampled MRI shown in each row. LIME uses a different repeated image and is invalid because of uint8 wraparound. The overlays are implementation diagnostics, not anatomical evidence.

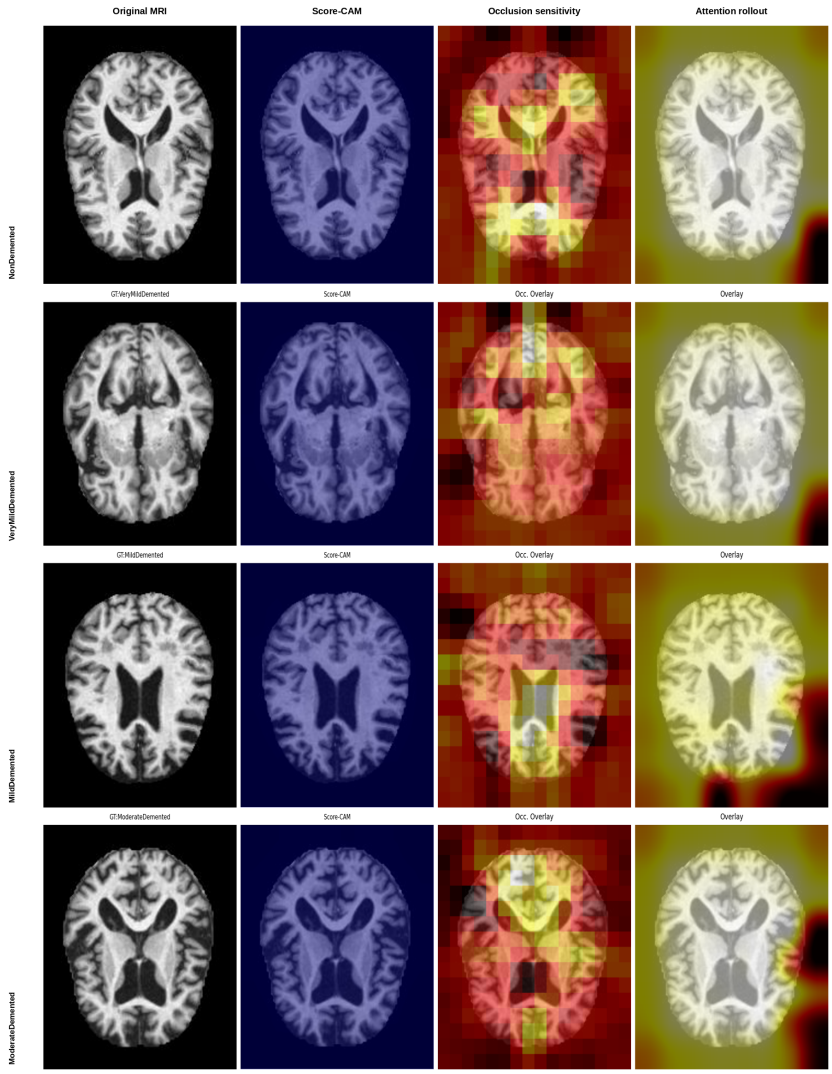


Fig. 14 Internal attribution implementation audit showing Score-CAM-style, occlusion and partial attention-rollout overlays. Score-CAM used at most 64 channels, and occlusion used 32×32 patches at stride 16. These approximate outputs provide no quantitative evidence of anatomical validity.

4.8 Computational Cost

The architecture has 99.63 million total and 64.94 million initially trainable parameters, identical across all four strategies (Table 16). Inference latency ranged from 52.38 ms to 57.28 ms over the notebook's 50 timed single-image passes; since all four share an architecture, that 4.90-ms spread should be treated as measurement variation. One float32 model payload was 380.07 MiB. With two clients receiving and returning a full payload, nominal bidirectional traffic is approximately 1,520.27 MiB per round and 30,405.43 MiB over 20 rounds, before protocol overhead. The recorded 2,444-MiB CUDA allocation is process-level because all four models remained resident during the batch-32 measurement; it is not a clean per-model memory estimate.

Table 16 Computational complexity and federated communication requirements

Strategy	Total params (M)	Trainable params (M)	Recorded CUDA alloc. (MiB)	Inference latency (ms)	Payload (MiB/transfer)
Centralized	99.63	64.94	2,444.0	52.38	N/A
FedAvg	99.63	64.94	2,444.0	54.82	380.07
FedProx	99.63	64.94	2,444.0	57.28	380.07
FedBN	99.63	64.94	2,444.0	55.44	380.07

The CUDA value is a process-level batch-32 allocation with four models resident. A two-client round entails four payload transfers, approximately 1,520.27 MiB before protocol overhead. Communication is not applicable to centralized training.

5 Discussion

5.1 What the Internal Results Establish

Internally, the framework achieved 95.13% balanced accuracy, 0.991 macro-AUC and 0.881 MCC, values within the range reported for comparable hybrid attention architectures in Tables 1 and 2 [11–13]. Considered alone, these results would suggest a successful federated framework.

The interpretation must be narrower for two reasons. The internal test partition shares a source distribution with the training clients, so it poses a task closer to interpolation within a known acquisition regime than to evaluation on a new source. More specifically, the training cohort carries no subject identifiers, so its split is image-level, and a slice-level release may place adjacent or near-duplicate slices on both sides. The internal partition may therefore be easier than a subject-disjoint partition, and 95% balanced accuracy on it is not evidence that the model would approach that value on unseen subjects. The OASIS test images come from identified OASIS subjects, but independence from the provenance-unknown training collection cannot be proven. The observed drop may combine acquisition shift with internal leakage; we cannot separate those contributions with the data available.

The apparent federated advantage over centralized training should not be interpreted as evidence of an aggregation benefit. The centralized and federated regimes differed in effective gradient updates and optimization schedule, and the federated routine constructed new ImageNet-initialized models instead of continuing from the centralized checkpoint. Moreover, the three aggregation strategies did not share an identically reset random head initialization. These confounders prevent a causal comparison of centralized and federated optimization. The paired McNemar test found no statistically significant difference in per-sample correctness between the final FedBN-labelled and FedAvg checkpoints ($p = 0.787$), but this does not demonstrate equivalence. The result is consistent with the shared client source and the limited implementation difference, because the FedBN-style rule excludes only one classifier layer's running buffers from averaging and does not maintain persistent personalized batch-normalization states.

5.2 The Transfer Failure

External macro-AUC ranged from 0.634 to 0.692, compared with 0.894 to 0.991 internally. Some one-versus-rest ranking signal therefore survived, but the fixed multiclass assignments did not transfer successfully. Table 13 makes this concrete: both evaluated models rarely predicted NonDemented, producing 65,684 and 55,807 disease false positives against 3 and 71 false negatives. With NonDemented representing 77.8% of the external collection, systematic over-assignment of disease labels resulted in accuracy of only 12% to 22%, despite above-chance ranking for several classes.

Table 12 establishes a large raw-image source difference: the KS statistic for per-image intensity standard deviation was 0.997. Because those statistics were calculated before the model transform, they do not quantify residual separation after z-scoring and cannot by themselves identify the mechanism of failure. They do show that the model was evaluated on a markedly different image source. Intensity normalization is not scanner harmonization, and a paper reporting the former should not imply the latter.

Two consequences follow for practice. Selection by internal macro-AUC would have chosen the FedBN-style checkpoint, yet the centralized model had the better external macro-AUC, so the internal criterion reordered the candidates. The FedBN-style model also made 9,711 high-confidence errors externally against 10 internally, so a deployment referring only low-confidence cases for expert review would have passed nearly ten thousand confident mistakes through untouched.

5.3 Calibration

On the centralized validation predictions, temperature scaling raised ECE from 0.348 to 0.390 and the Brier score from 0.136 to 0.149. Being monotone, it could not harm ranking; the recorded result shows only that this fitted scalar did not improve the chosen calibration measures on those predictions. Per-class reliability deviations ranged from 0.203 to 0.450. A separate calibration-test partition and external probability analysis would be needed before assigning a clinical meaning to confidence values. Vector or Dirichlet scaling, or calibration fitted and evaluated with target-site data,

are appropriate next steps. Only one of the forty studies in Tables 1 and 2 reports calibration [18].

5.4 Comparison with Related Work

The closest comparator is Abubakr et al. [14], who report 94.7% federated accuracy internally and 95.0% externally, a difference of 0.3 percentage points. Our internal figure is comparable at 92.74%, whereas our external figure is 22.11%. This contrast does not by itself establish greater architectural robustness in their framework. Their task was binary rather than four-stage, and their external set was another public Kaggle-derived collection rather than a differently acquired cohort. External evaluations vary substantially in difficulty; the relationship between the training and evaluation sources must therefore be described explicitly.

Yao [15] reports 98.3% from eight clients formed by partitioning one 6,400-image collection. Our federated setup shares that limitation. When every client draws from one source, aggregation is not required to reconcile different acquisition characteristics; consequently, strong performance in such a simulation does not establish the cross-site generalization that federated learning is intended to support.

5.5 Limitations

In rough order of how much they constrain the conclusions. The training cohort has no subject identifiers, so its split is image-level and slice-level leakage cannot be excluded or measured; every internal figure, including cross-validation, is an upper bound. Subject-level independence between the two sources is likewise not established, so any overlap would make the external results optimistic. The federation is simulated from one cohort, so the FedAvg, FedProx and FedBN-style comparison was conducted without the site heterogeneity those methods address. The FedBN-style implementation excludes running buffers from averaging but does not maintain persistent client-specific batch-normalization state. The centralized-versus-federated gap is unexplained and should not be cited as evidence for an aggregation benefit.

The executed objective also differs from its intended design. The supervised contrastive routine produced a non-finite value after masking and its guard replaced the term with zero, so the reported experiments evaluate soft-target cross-entropy and focal loss rather than a functioning three-term objective. The class-weighted cross-entropy object and label-smoothing setting were bypassed by the soft-target path. The Optuna pruner was configured but never invoked, and the search ranges differed from earlier notebook summaries; Table 4 now reports the executed ranges.

The attribution outputs are implementation diagnostics rather than validated XAI evidence. The Grad-CAM++ and Score-CAM routines are approximations, the attention rollout is partial, the LIME wrapper corrupts perturbed intensities through uint8 wraparound, and SHAP yielded no retained output. Completed blinded clinical ratings, quantitative faithfulness, atlas registration and cross-method agreement are unavailable. The architecture-baseline and component-ablation results were obtained from single runs, so their uncertainty across random seeds is unknown and small differences should be interpreted cautiously. The complete model led all reported internal

measures and most external decision metrics, but the model without CBAM achieved a slightly higher external macro-AUC. The CBAM module operates on a tensor of unit spatial extent, so its spatial branch reduces to a scalar gate; placing CBAM on a genuine spatial feature map before pooling would be the more natural design and was not tested.

Further limitations concern measurement and metadata. The raw shift statistics use 300 sampled images per source and do not quantify residual shift after preprocessing. The reported CUDA allocation is process-level rather than isolated per model, and the saved communication value is a single payload rather than full per-round traffic. FLOPs, complete training time, CPU memory and deployment performance were not measured. ModerateDemented holds only 9 internal and 488 external images, external calibration was not quantified, classification is on 2D slices rather than volumes, and no acquisition metadata were available, so the raw shift could be measured but not attributed. The corruption analysis covered four synthetic perturbations but did not test reduced training data or missing slices.

Several additional artifacts were exploratory or incomplete and were therefore not treated as findings. The uncertainty output did not provide complete external validation or a clinically validated referral threshold. The recorded age and sex fairness variables and the kappa values were simulated, ethnicity was unavailable, and the generated clinical-review panels contained no completed expert ratings. The t-SNE, UMAP and PCA plots were qualitative and lacked a matched baseline or quantitative separation analysis. These outputs cannot support demographic fairness, clinical agreement, uncertainty-based deployment or representation-quality claims.

5.6 Implications for Practice

Three recommendations follow as reporting practice. Report the distance between training and evaluation sources, not just the fact of external validation: simple per-image statistics can show whether an external result reflects a genuine change of source. Report calibration alongside discrimination and reserve separate data for fitting and evaluating calibration; our validation result showed that scalar temperature scaling did not improve the recorded measures, while the external high-confidence error count was nearly three orders of magnitude higher than its internal value. Finally, treat synthetic corruption robustness and cross-source robustness as different claims, since the synthetic corruptions changed accuracy by at most 2.42 percentage points while the source change reduced it by 56.50 to 72.33 percentage points.

For the framework itself, the productive next steps target the failure actually observed: harmonization or domain adaptation at the input stage, threshold or calibration adaptation using a small labelled sample from the target site, a subject-labelled training cohort so the internal result can be measured rather than bounded, and a genuinely multi-institutional federation in which the clients differ.

6 Conclusion

We evaluated a dual-channel framework combining ConvNeXt-Small and Swin-V2-S branches with CBAM refinement under centralized training, FedAvg, FedProx and a

FedBN-style rule. The federated experiments used two simulated clients drawn from a single imaging source. The executed objective comprised soft-target cross-entropy and focal loss because the configured supervised contrastive component was numerically inactive. On the 951-image internal test partition, the FedBN-style checkpoint achieved 95.13% balanced accuracy, a macro-AUC of 0.991 and an MCC of 0.881. Its per-sample error pattern differed significantly from that of the centralized checkpoint.

Performance did not transfer to the 86,437-image OASIS-1-derived external collection. Across the four strategies, external accuracy ranged from 12.06% to 22.11% and macro-AUC from 0.634 to 0.692, corresponding to accuracy reductions of 56.50 to 72.33 percentage points relative to the internal test results. The centralized and FedBN-style checkpoints rarely predicted the majority NonDemented class, and the FedBN-style model produced 9,711 high-confidence external errors compared with 10 internally. In reproducible samples of 300 raw images per source, the Kolmogorov–Smirnov statistics were 0.967 for per-image mean intensity and 0.997 for per-image intensity standard deviation, with $\mathrm{MMD}^2 = 0.534$. These statistics demonstrate a large sampled raw-source difference but do not measure residual shift after preprocessing. Temperature scaling on the same centralized validation partition increased expected calibration error from 0.348 to 0.390 rather than improving it.

The cross-source failure is therefore the principal finding. Internal macro-AUC and external macro-AUC favoured different checkpoints, robustness to four synthetic corruptions did not anticipate the source-related degradation, and a referral rule based only on low confidence would not have captured the many confident external errors. Because the training collection lacks subject identifiers, its internal split is image-level and may contain related or near-duplicate slices across partitions. The available evidence cannot separate the effects of possible internal leakage, source shift and model design, nor does the same-source federation establish robustness across institutions.

Future evaluation should prioritize subject-disjoint training data, genuinely heterogeneous institutional clients, measurement of source differences after preprocessing, external calibration and quantitatively validated attribution methods. Reporting source characteristics, calibration and confidence-stratified errors alongside discrimination metrics is necessary to distinguish strong same-source performance from evidence of clinically relevant generalization.

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Declarations

Funding. No external funding was received for this work.

Competing interests. The authors declare no competing interests.

Data availability. No new datasets were generated for this study. The 6,400-image training collection was obtained from the OriginalDataset partition of the Augmented Alzheimer MRI Dataset [41]. The external collection was obtained from the OASIS Alzheimer's Detection derivative [42]; the underlying OASIS-1 collection and its documentation are described by Marcus et al. and the OASIS project website [43, 44].

All resources are publicly accessible, although access may require registration and acceptance of the respective provider terms.

Code availability.

Author contributions.

Ethics approval. This study used publicly available, de-identified imaging collections and did not require institutional ethics approval.

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